

Assessment 1

Sunday, 30 July 2023 00:17

[illegible]

Question 4 (Assessment 1)

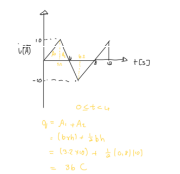
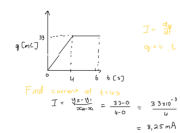
$t = 2 \text{ hours}$
 $E = 240 \text{ kJ}$
 $V = 20 \text{ V}$
 $I = ?$

$\frac{W}{t} = VI$
 $I = \frac{W}{Vt}$
 $= \frac{240 \times 10^3}{(20)(7200)}$
 $= 1,67 \text{ A}$

Question 3 (Assessment 1)

$q = -20 \text{ nC}$
 $t = 3 \mu\text{s}$
 $w = 40 \text{ mJ}$
 $J = ?$

$$I = \frac{q}{t}$$
$$= \frac{-20 \times 10^{-9}}{3 \times 10^{-6}}$$
$$= -6.67 \text{ mA}$$



Question 1 (Assessment 1)


The graph shows a current i (A) versus time t (s). The current starts at 0 at $t=0$, reaches a maximum value of 1.25 A at $t=27$ ms, and then decays. A point on the curve is marked at $t=7$ ms with a current of 0.4 A.

Find I at $t = 7$ ms
 0.4

Find I at $t = 45$
 $I = \frac{2.5 \times 10^{-3}}{0.4 - 0} = \frac{2.0 \times 10^{-3}}{0.4}$
 $= 7.5$ mA

Question 5 (Assessment 1)

1. $V_t = \frac{d\phi}{dt} V$ $\phi(t) = A e^{-\omega t}$ [V]

2. $i(t) = 20 e^{-\omega t} \text{ mA}$

3. $V = - \frac{q}{C} e^{-\omega t} \times 10^{-3} \text{ V}$ [V]

4. $P = V_i I_i$

5. $= (-8 e^{-\omega t} \times 10^{-3}) (20 e^{-\omega t})$

6. $= -16 e^{-2\omega t} \times 10^{-3} \text{ W}$

7. $= -16 e^{-2\omega t} \times 10^{-3}$

8. $\therefore A = -16$

9. $B = -8$

10. $= -16 e^{-2\omega t} \mu \text{ W}$

Question 7 (Assessment 1)

V_1

R_1

R_2

V_x

i_1

$0.5i_1$

P_1

$P_1 = -[V_x \times i_1]$

$= -[2 \times 100]$

$= -200 \text{ W}$